

Skills Summary

Binomial Coefficients and Expansion Structure

Use this reference while completing your worksheet or when studying.

Skill 1 — Expanding a binomial power

The pattern: $(A + B)^n$ has $n + 1$ terms. The powers of A fall from n to 0 while the powers of B rise from 0 to n , and the coefficients are row n of Pascal's triangle (1, 3, 3, 1 for $n = 3$; 1, 4, 6, 4, 1 for $n = 4$).

Sign and coefficient travel with the letter. If B is negative the signs alternate; if A is $2x$, then A^3 is $8x^3$, not $2x^3$.

Example: Expand $(x + 2)^3$.

Step 1. The exponent is 3, so the coefficients are row 3 of the triangle and the powers of 2 climb as the powers of x fall — the whole expansion can be written straight down.

Step 2. 1, 3, 3, 1 against $2^0, 2^1, 2^2, 2^3$: $x^3 + 3x^2(2) + 3x(4) + 8$.

$$\Rightarrow x^3 + 6x^2 + 12x + 8$$

Try it: Expand $(x - 1)^4$.

check: $1 + x^4 - 4x^3 + 6x^2 - 4x + 1$

Skill 2 — Picking out one named term

Do not expand. The term at index k of $(A + B)^n$ is $\binom{n}{k}A^{n-k}B^k$, so a question about one power of x fixes k and the rest is arithmetic.

Find k from the power you want. If $A = x$ and you want x^m , then $n - k = m$, so $k = n - m$. Index k starts at 0, so the third term has $k = 2$.

Example: Find the coefficient of x^2 in $(x + 3)^4$.

Step 1. Only one term carries x^2 , so the index is forced: $n - k = 2$ with $n = 4$ gives $k = 2$, and nothing else needs computing.

Step 2. $\binom{4}{2}x^23^2 = 6 \cdot 9 \cdot x^2$

$$\Rightarrow \text{coefficient} = 54$$

Try it: Find the coefficient of x^2 in $(x + 4)^4$.

check: 96

Skill 3 — Reading a binomial coefficient

The formula: $\binom{n}{k} = \frac{n!}{k!(n-k)!}$, which counts the ways of choosing k of the n factors to contribute their second term.

Symmetry: $\binom{n}{k} = \binom{n}{n-k}$, because choosing k to take is the same as choosing $n - k$ to leave. That is the mirror symmetry you can see in every row of Pascal's triangle.

Example: Compute $\binom{7}{2}$.

Step 1. The formula counts choices, so only the top two factors of $7!$ survive against $2!$ — there is no need to write out either factorial in full.

Step 2. $\frac{7 \cdot 6}{2} = \frac{42}{2}$

$\Rightarrow$ **21** (and by symmetry $\binom{7}{5}$ is the same number)

Try it: Compute $\binom{8}{3}$.

check: 56

Skill 4 — Summing across an expansion

Substituting $x = 1$ collapses an expansion to its coefficient total, because every power of 1 is 1. For $(1 + x)^n$ that total is 2^n — one row of Pascal's triangle adds to 2^n .

Those totals themselves form a geometric sequence with ratio 2, so a sum of several rows is an ordinary geometric series: $a_1(r^m - 1)/(r - 1)$.

Example: Evaluate $\sum_{k=0}^4 \binom{4}{k}$.

Step 1. The sum runs over a whole row of Pascal's triangle, so it is the coefficient total of $(1 + x)^4$ — the substitution shortcut applies.

Step 2. Row 4 is 1, 4, 6, 4, 1, and $1 + 4 + 6 + 4 + 1 = 2^4$.

$\Rightarrow$ **16**

Try it: Evaluate $\sum_{n=0}^4 3^n$.

check: 121